

INTERNSHIP REPORT

On

**(AI/ML)ANPR Model for Smart Parking
System**

By

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Regd. No.BU22CSEN0300538

PARKNSECURE

INTN3444

(Duration: 05/05/2025 to 20/06/2025)



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Gandhi Institute of Technology and Management

(DEEMED TO BE A UNIVERSITY)

BENGALURU, KARNATAKA, INDIA

SESSION:2022-2026

Certificate Issued After Internship (only for Students Completed from Industries)

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ParkNSecure Private Limited Internship Certificate

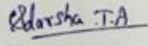
To Whom It May Concern,

This is to certify that **C H UDAY KUMAR** has successfully completed a 45-day internship in the domain of **AI/ML** at **ParkNSecure Private Limited, Bengaluru**, from 5 May 2025 to 20 June 2025.

During this period, the intern actively contributed to real-world projects and participated in a collaborative, fast-paced startup environment. Their role involved hands-on work, continuous learning, and exploration of innovative solutions aligned with **ParkNSecure Private Limited's** mission to revolutionize smart parking with AI-driven technologies.

This internship experience was part of our structured learning and prototyping initiative, blending product development with entrepreneurial thinking.

We appreciate their involvement and wish them success in all future endeavours.

For ParkNSecure Private Limited

Director

Adarsha T A
Founder & CEO,
ParkNSecure Private Limited.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
GITAM SCHOOL OF TECHNOLOGY



CERTIFICATE

This is to certify that the mini project report entitled “**ANPR Model for Smart Paring System**” is a bonafide record of work carried out by **C H UDAY KUMAR (BU22CSEN0300538)** submitted in partial fulfillment of requirement for the award of degree of **Bachelor of Technology in Computer Science and Engineering**.

Nitturu Asha Jyothi
Assistant Professor
Department of AI & DS,
GST, Bengaluru

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of the people who made it possible, whose consistent guidance and encouragement crowned our efforts with success.

We consider it our privilege to express our gratitude to all those who guided us in the completion of the project.

We express our gratitude to Director Prof. **Basavaraj Gundappa Katageri** for having provided us with the golden opportunity to undertake this project work in their esteemed organization.

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C H UDAY KUMAR

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Abstract

This project introduces an Automatic Number Plate Recognition (ANPR) system designed to automatically detect and recognize vehicle license plates using a combination of deep learning and Optical Character Recognition (OCR) technologies. The system is particularly tailored for Indian license plate formats, which vary in structure, size, and font style across different states, presenting a unique challenge that this solution effectively addresses.

The ANPR system operates in real-time, enabling license plate recognition directly from live webcam streams, recorded video files, or still images. The pipeline includes several key components: vehicle detection, license plate localization, plate image enhancement, character segmentation, and finally, OCR-based text extraction. Deep learning models, such as Convolutional Neural Networks (CNNs), are used for object detection (vehicles and plates), ensuring high accuracy even in challenging conditions such as varying lighting, occlusion, and complex backgrounds.

One of the core strengths of this solution is its modular architecture. Each component of the system is developed as a standalone module, allowing for flexible customization, easy maintenance, and future upgrades. This modularity makes the system suitable for integration into broader applications, such as automated toll collection, traffic law enforcement, parking management, and secured entry access systems.

The system prioritizes accuracy and performance, aiming to minimize false positives and improve recognition rates for Indian number plates, which often deviate from standardized formats. Additionally, it is optimized for computational efficiency, enabling deployment on edge devices or low-resource environments if needed.

In summary, this ANPR system provides a robust, scalable, and adaptable solution for real-time vehicle identification, with a strong focus on practical deployment in Indian traffic and infrastructure scenarios.

INTRODUCTION

Overview of the Problem Domain

With the rapid growth of urban areas and vehicle ownership, cities across the globe—especially in countries like India—are facing increasing challenges in traffic management, road safety, and secure access control. Manual monitoring of vehicles at entry and exit points is time-consuming, error-prone, and inefficient. Traditional methods for vehicle identification and documentation require human intervention, which leads to delays, inconsistencies, and scalability issues.

In particular, facilities such as toll plazas, residential communities, government offices, parking lots, and commercial complexes need a more reliable and automated way of identifying and recording vehicle entries. The lack of digitization in this domain leads to:

- Poor vehicle tracking and record maintenance
- Inefficient parking and access control
- Security vulnerabilities
- Resource-intensive operations

To address these issues, Automatic Number Plate Recognition (ANPR) systems offer a promising solution by automating the detection and recognition of license plates using cameras and computer vision. These systems, when implemented correctly, can provide real-time data, enhance security, streamline parking, and improve overall traffic efficiency.

Objective of the Internship

The objective of this internship was to contribute to the development and optimization of an Automatic Number Plate Recognition (ANPR) system, with a particular focus on detection accuracy, OCR performance, and system integration. The system was developed as a modular solution suitable for Indian road conditions and license plate formats.

Project Scope

The scope of the project was centered around building and validating an end-to-end ANPR pipeline, which included:

- Vehicle and license plate detection using YOLOv8

- Character segmentation and recognition via OCR
- Data extraction and storage for future logging or access control integration
- Modular architecture design for future extensibility and deployment
- Real-time capabilities using efficient processing strategies for smooth input handling

Importance

The ANPR system developed as part of this project is highly relevant to modern smart city initiatives and traffic management strategies. Key benefits include:

- Enhanced security through automated vehicle logging
- Faster access control in residential or commercial facilities
- Reduction in manual labor and associated errors
- Scalability for city-wide surveillance and monitoring systems
- Data-driven insights for law enforcement and urban planning

This internship provided practical exposure to building a real-world computer vision system that addresses pressing urban challenges.

Tools & Technologies

The following tools and technologies were used during the course of the internship:

- YOLOv8 – Real-time object detection
- Tesseract OCR, EasyOCR – Text recognition engines
- OpenCV, NumPy, Pillow – Image preprocessing and manipulation
- Python – Core programming language
- Jupyter Notebooks – For experimentation and prototyping
- Git & GitHub – Version control and collaboration
- VS Code – Development environment

Company Overview

ParkNSecure is a technology-driven company focused on providing innovative and smart solutions for urban infrastructure, with a particular emphasis on intelligent parking systems. Their flagship product, SecurePark, is designed to address the rising challenges of urban traffic congestion, inefficient parking management, and the growing demand for digital solutions in smart cities. The company offers end-to-end solutions integrating hardware (such as sensors and barriers) with software components (like mobile and web apps) to deliver a seamless, secure, and real-time parking experience to users.

Their services are aligned with sustainable urban mobility goals, and their solutions are deployed in various public and private sector environments. With a strong commitment to innovation, user experience, and data security, ParkNSecure plays a key role in shaping the future of urban mobility through intelligent automation and real-time data integration. ParkNSecure's development teams adopt Agile methodologies and modern technology stacks to ensure scalability, maintainability, and faster deployment cycles.

This approach allows them to respond quickly to evolving user needs and continuously improve their product offerings. By combining IoT devices, AI-driven analytics, and cloud-based infrastructure, ParkNSecure empowers cities to optimize parking operations, enforce compliance, and collect meaningful data for long-term urban planning. Their mission aligns with the broader vision of creating smarter, safer, and more efficient urban environments.

Project Overview

The Automatic Number Plate Recognition (ANPR) system developed during this project is a modular, scalable, and real-time computer vision application. It is designed to detect and recognize vehicle license plates from various input sources, with a strong emphasis on Indian license plate formats. The system integrates state-of-the-art deep learning and OCR technologies and is structured into well-defined modules to ensure flexibility, maintainability, and ease of deployment. Below is a detailed breakdown of the major components:

1. Real-time License Plate Detection using YOLOv8

At the core of the system is a YOLOv8 (You Only Look Once, Version 8) deep learning model, trained specifically for license plate detection. YOLOv8 is known for its high accuracy and fast inference speed, making it suitable for real-time applications. This module is capable of:

- Detecting license plates in live webcam feeds, video files, or static images
- Handling diverse lighting conditions, partial occlusions, and various viewing angles
- Operating efficiently on both CPU and GPU setups

The detection model was trained using a dataset formatted in YOLOv8 standards, ensuring smooth compatibility with the Ultralytics YOLO training pipeline.

2. License Plate Text Extraction using OCR

Once a license plate is detected and cropped, the system applies OCR (Optical Character Recognition) to extract alphanumeric text. Two open-source OCR engines are integrated:

- Tesseract OCR – Well-established, customizable, supports language training and post-processing
- EasyOCR – Deep learning-based, supports multilingual recognition and performs well on noisy data

Additional processing steps improve recognition accuracy:

- Image enhancement (grayscale conversion, thresholding, denoising)
- Character segmentation (if needed)
- Post-processing using regex patterns to match Indian plate formats and correct common OCR errors

This pipeline ensures that the extracted text is accurate, clean, and ready for downstream tasks such as validation or storage.

3. Dataset Management & Model Training Pipeline

To develop a robust detection model, the system uses a dataset sourced from Roboflow, containing over 3,800 annotated images of vehicles with labeled license plates in YOLOv8 format. Key aspects of this module include:

- Automated data preprocessing and augmentation (flipping, scaling, brightness adjustment)
- Structured training using the `train_anpr_model.py` script, which wraps the Ultralytics YOLO training API
- Periodic validation and testing to monitor overfitting and generalization performance
- Option to fine-tune the model with additional data or improve it through active learning loops

This training setup ensures adaptability to new environments and vehicle types, including those unique to different regions of India.

4. Prebooked Plate Verification for Access Control

To support smart access control, the system includes a prebooked vehicle verification feature. After extracting a plate number, the system checks it against a list of authorized vehicles, stored in a local or remote database (e.g., CSV, JSON, or API-based storage). This enables real-time decisions such as:

- Granting or denying access to parking lots or gated communities
- Logging unauthorized attempts
- Sending alerts or updating dashboards

This feature is particularly useful for applications in secure parking management, corporate campuses, and residential areas.

Flow Chart

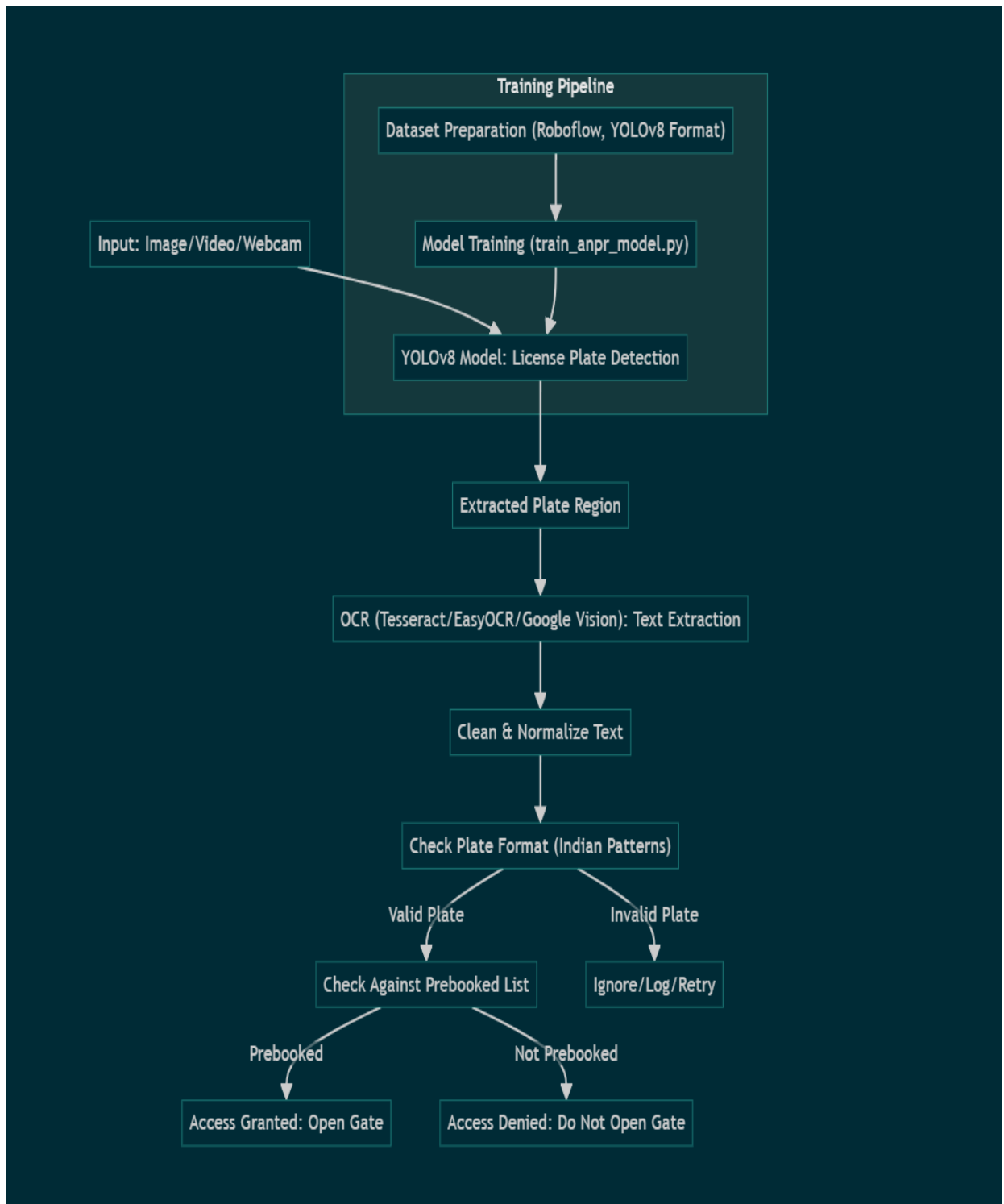


Fig:-4.1 Flow chart of the complete project

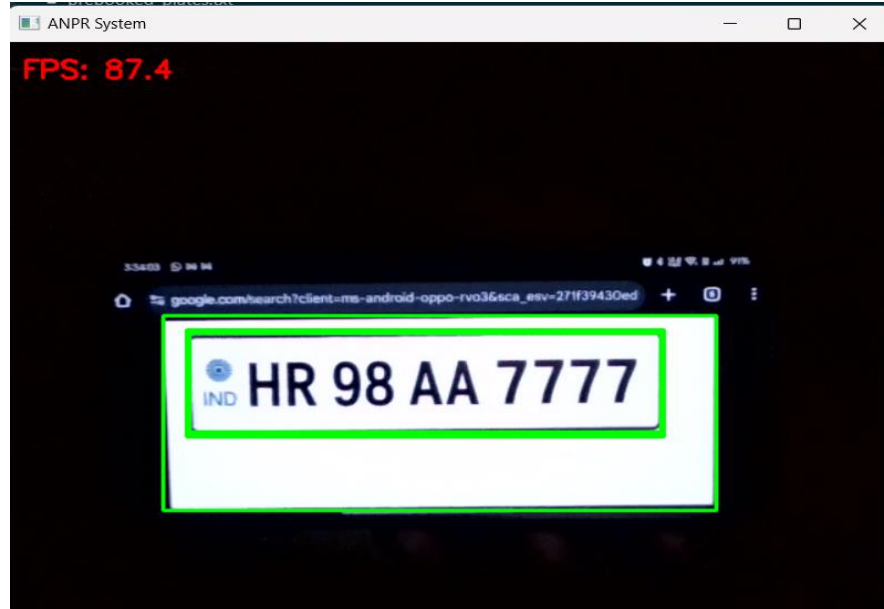


Fig:-4.2 ANPR System

Description:-The above figure represented that the trained ANPR Model is Detecting the license plate and capturing the numbers in the number plate and sending signals to barriers to open gate or not.

```

Detected plate: KA19EQ0001
✓ Prebooked. OPEN GATE.

Detected plate: KA19EQ0001
✓ Prebooked. OPEN GATE.

Detected plate: KA19EQ0001
✓ Prebooked. OPEN GATE.

Detected plate: HR98AA7777
✗ Not prebooked. DON'T OPEN GATE.

Detected plate: HR98AA7777
✗ Not prebooked. DON'T OPEN GATE.

Detected plate: HR98AA7777
✗ Not prebooked. DON'T OPEN GATE.

Detected plate: HR98AA7777
✗ Not prebooked. DON'T OPEN GATE.
  
```

Fig 4.3:-output

Output:-The below figure shows that when an detected plate is in the prebooked list , then it gives the permission to open the gate or else it will not allow to open the gate.

Tasks & Work Breakdown

The ANPR system development was structured into distinct tasks, each contributing to a specific part of the overall pipeline. This modular breakdown helped ensure clarity of responsibilities, better debugging, and smooth integration between components. Below is a detailed explanation of the key tasks completed during the project.

1. Dataset Preparation and Annotation for YOLOv8

The first critical step was acquiring and preparing a high-quality dataset. This involved:

- Sourcing data from platforms like Roboflow, which provided a diverse set of vehicle images with labeled license plates
- Performing additional manual annotation (if necessary) using tools like Roboflow Annotate or LabelImg to define bounding boxes around plates
- Ensuring all annotations adhered to the YOLOv8 format, which includes class ID and bounding box coordinates normalized relative to image dimensions
- Organizing the dataset into training, validation, and test splits to enable proper model evaluation
- Applying data augmentation techniques (e.g., brightness adjustment, blurring, rotation) to increase model robustness

This foundational task ensured the detector had enough representative examples to generalize well across various environments and lighting conditions.

2. Training the YOLOv8 Model for License Plate Detection

After dataset preparation, the next step involved training a custom YOLOv8 model tailored for license plate detection:

- Configuring Ultralytics YOLOv8 training parameters such as batch size, epochs, learning rate, and image size

- Using the `train_anpr_model.py` script to initiate and monitor the training process
- Tracking metrics like precision, recall, and mAP (mean Average Precision) using built-in tools
- Running validation and testing phases to evaluate the model's generalization ability
- Exporting the trained model (.pt file) for integration into the detection pipeline

This task resulted in a high-performance detector capable of accurately localizing plates in real-world scenarios.

3. Developing Detection Scripts for Real-Time and Batch Processing

With a trained model in place, custom detection scripts were developed to handle different input modes:

- `api.py`: Designed for real-time streaming input, capturing frames from a webcam and performing live detection and OCR
- `new.py`: Used for testing images and video files, useful during offline debugging and performance benchmarking
- Integrated OpenCV for video capture and image manipulation
- Implemented real-time optimizations such as frame skipping, resizing, and threading to ensure smooth performance even on mid-range hardware

These scripts enabled flexible deployment, whether in a lab setting, roadside setup, or integrated with existing camera infrastructure.

4. OCR Integration and Text Post-Processing

Once plates were detected and cropped, the next task involved extracting readable text using OCR. This step included:

- Evaluating and integrating Tesseract and EasyOCR to handle character recognition

- Preprocessing the cropped plate image (grayscale conversion, thresholding, noise reduction) to improve OCR accuracy
- Implementing text cleaning logic using Python (e.g., removing special characters, fixing spacing issues)
- Applying regular expressions (regex) to validate the recognized string against expected Indian number plate formats
- Creating fallback mechanisms where OCR confidence was low (e.g., switching engines or retrying with enhanced images)

This ensured a reliable and structured output from raw license plate images, even in noisy or challenging scenarios.

5. Prebooked Plate Verification for Access Control

To make the system useful for real-world applications like gated entry or automated parking, a prebooked verification module was added. This involved:

- Maintaining a list or database of authorized license plate numbers, stored in formats such as CSV or JSON
- Comparing the OCR output against the whitelist using string matching algorithms
- Returning a real-time access decision (e.g., “Access Granted” or “Unauthorized Vehicle”)
- Logging all recognition events along with timestamps for audit and analysis purposes

This feature enabled smart automation in parking lots, secured campuses, and residential societies.

6. System Testing and Real-Time Optimization

Finally, the full pipeline was tested across a variety of input types and scenarios:

- Webcam streams under different lighting and angles

- HD and low-resolution video files with moving vehicles
- Static images with occluded, skewed, or distant plates
- Stress testing to check latency, frame rate stability, and detection accuracy

Real-time optimization involved:

- Reducing image resolution for faster inference
- Batching OCR requests or introducing cooldowns between reads
- Utilizing GPU acceleration if available
- Handling failure cases gracefully (e.g., if no plate is detected)

These tests ensured the system met performance expectations for live deployment environments.

Tools & Technologies Used

1. Python 3.8+

- Versatile scripting language used for developing the entire ANPR pipeline, offering strong support for machine learning, computer vision, and web frameworks.
- Extensive libraries and community support make it ideal for rapid prototyping and integration of AI and OCR components.

2. YOLOv8 (Ultralytics)

- State-of-the-art object detection model that provides fast and accurate bounding box predictions for license plates, essential for real-time performance.
- Pretrained model support and transfer learning capabilities allow fine-tuning on custom datasets, like Indian license plate images, for improved accuracy.

3. EasyOCR and Tesseract

- EasyOCR is a deep learning-based OCR engine that supports multilingual recognition and performs well on noisy or non-standard images.
- Tesseract is an open-source OCR engine by Google, known for its accuracy with clean, well-formatted text and its support for custom training and text pattern recognition.

4. OpenCV

- Used for image and video processing, such as reading frames, cropping plate regions, and applying filters (e.g., grayscale, thresholding).
- Enables real-time video stream handling, making it crucial for implementing live ANPR functionality using webcam or CCTV inputs.

5. NumPy

- Provides efficient array operations used throughout the pipeline for manipulating image data and bounding box coordinates.
- Supports numerical computations that are essential for tasks like image normalization and post-processing of model outputs.

6. Torch and Torchvision

- Core deep learning libraries used for model training, evaluation, and inference, particularly with YOLOv8 (built on PyTorch).
- Torchvision provides image utilities, datasets, and transformation functions to streamline the model development process.

7. Flask (Optional)

- Lightweight web framework used for exposing the ANPR pipeline as an API, enabling easy integration with frontends or external systems.
- Supports real-time HTTP endpoints, allowing vehicle recognition to be accessed remotely (e.g., via a mobile app or dashboard).

8. Pillow, PyYAML, tqdm

- Pillow is used for basic image manipulation tasks like resizing, cropping, and saving processed plate images.
- PyYAML helps with reading configuration files (e.g., model settings, dataset paths), improving maintainability and flexibility.
- tqdm provides progress bars during long-running tasks like training or batch processing, enhancing developer feedback and debugging.

9. CUDA GPU (Recommended)

- Accelerates deep learning tasks, significantly reducing training and inference time when using YOLOv8 and OCR models.
- Enables real-time processing, especially for high-resolution video streams or multiple concurrent detections, which would be slow on CPU alone.

Results

The developed ANPR system successfully meets its core objectives, demonstrating both high accuracy and real-time performance. When deployed on a GPU-enabled system, the pipeline is capable of processing video feeds at 20 to 40 frames per second (FPS), enabling seamless real-time detection and recognition of license plates. Even on CPU-only systems, the performance remains functional, with speeds ranging between 1 to 5 FPS, making it suitable for lower-end or embedded applications where real-time speed is not critical.

The YOLOv8 detection model was trained on a carefully curated dataset of 3,855 annotated images sourced and prepared via Roboflow. Training and evaluation results indicate strong model performance in terms of precision, recall, and mean average precision (mAP). These metrics, detailed in the file `model/anpr_model/results.csv`, show that the model generalizes well across various real-world conditions, including differences in lighting, background clutter, and vehicle orientation.

The system demonstrates robust performance specifically on Indian license plates, which often vary in font, size, and formatting across different states. It handles these variations effectively, with OCR modules (Tesseract and EasyOCR) working in tandem to provide reliable text recognition. The solution also supports both cloud-based OCR pipelines for scalable deployments and local OCR processing for privacy-sensitive or offline use cases.

Another significant result of the project is its modular architecture, which ensures flexibility and ease of adaptation. The detection and recognition components are designed as interchangeable modules, allowing developers to swap models, retrain on different datasets, or integrate support for other regional formats and languages without rewriting the entire pipeline. This makes the solution highly extensible for broader use cases, including international deployments or specialized applications like commercial fleet management and law enforcement.

In summary, the ANPR system delivers accurate and efficient license plate recognition under real-world conditions, validated through quantitative metrics and practical testing across diverse input sources.

CONCLUSION

This ANPR project successfully demonstrates a practical and effective solution for automated vehicle license plate recognition by combining modern deep learning techniques with robust OCR technologies. By leveraging YOLOv8 for license plate detection and integrating OCR engines like Tesseract and EasyOCR for text recognition, the system is capable of delivering accurate and real-time results across a variety of environments and input types.

The modular and scalable architecture of the system makes it highly adaptable to a range of real-world applications, including access control systems, smart parking management, traffic enforcement, and urban surveillance. Its ability to handle Indian license plate formats, which are known for their variability, highlights the system's robustness and practical relevance in regional deployments.

Furthermore, the project emphasizes flexibility, allowing for easy extension to support additional countries or languages, custom database integrations, and frontend interfaces. The solution's support for both cloud-based and offline operation adds to its deployment versatility—accommodating environments with limited connectivity or strict data privacy requirements.

Looking ahead, future enhancements could focus on increasing OCR accuracy through better preprocessing, language model integration, or advanced AI-based character segmentation. There is also potential for integrating the system with cloud analytics platforms, real-time dashboards, or IoT-based edge devices to further enhance performance, scalability, and data-driven decision-making.

In conclusion, this project lays a solid foundation for a reliable and production-ready ANPR system, offering both technical depth and practical impact in the growing domain of smart mobility and intelligent transportation systems.